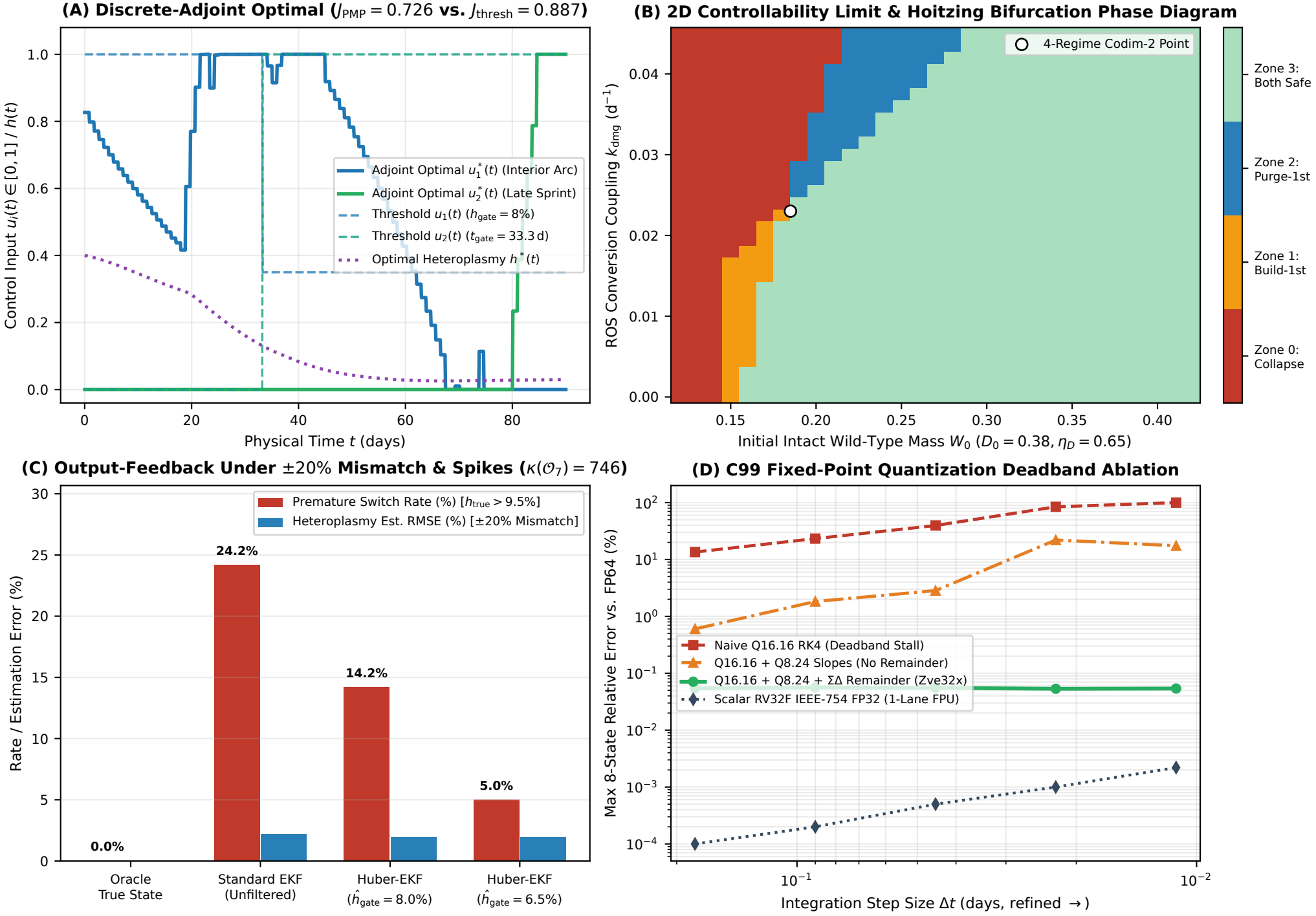

(A) Discrete-Adjoint Optimal ($J_{PMP} = 0.726$ vs. $J_{thresh} = 0.887$)
Control Input $u_i(t) \in [0, 1]$ / $h(t)$
Physical Time t (days)
Adjoint Optimal $u_1^*(t)$ (Interior Arc)
Adjoint Optimal $u_2^*(t)$ (Late Sprint)
Threshold $u_1(t)$ ($h_{gate} = 8\%$)
Threshold $u_2(t)$ ($t_{gate} = 33.3$ d)
Optimal Heteroplasmy $h^*(t)$
(B) 2D Controllability Limit & Hoitzing Bifurcation Phase Diagram
ROS Conversion Coupling k_{dmg} (d^{-1})
Initial Intact Wild-Type Mass W_0 ($D_0 = 0.38$, $\eta_D = 0.65$)
4-Regime Codim-2 Point
Zone 3: Both Safe
Zone 2: Purge-1st
Zone 1: Build-1st
Zone 0: Collapse
(C) Output-Feedback Under ±20% Mismatch & Spikes ($\kappa(\mathcal{O}_7) = 746$)
Rate / Estimation Error (%)
Premature Switch Rate (%) [$h_{true} > 9.5\%$]
Heteroplasmy Est. RMSE (%) [±20% Mismatch]
24.2%
14.2%
5.0%
0.0%
Oracle True State
Standard EKF (Unfiltered)
Huber-EKF ($\hat{h}_{gate} = 8.0\%$)
Huber-EKF ($\hat{h}_{gate} = 6.5\%$)
(D) C99 Fixed-Point Quantization Deadband Ablation
Max 8-State Relative Error vs. FP64 (%)
Integration Step Size Δt (days, refined →)
Naive Q16.16 RK4 (Deadband Stall)
Q16.16 + Q8.24 Slopes (No Remainder)
Q16.16 + Q8.24 + ΣΔ Remainder (Zve32x)
Scalar RV32F IEEE-754 FP32 (1-Lane FPU)